

MATH 3210-004 PSet 5

Specification

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1 Background

The common theme of all problems in this problem set is the convergence of a sequence of functions. The first problem focuses on uniform convergence and function spaces in the concrete setting of functions of the unit interval $[0, 1]$, while the remaining three problems develop an integration theory, namely integration in the sense of Bourbaki. Even compared to Riemann integration, Bourbaki integration is rather limited in terms of the kinds of real valued functions it admits as integrable¹. Still, the underlying mechanism of this integration theory is uniform convergence of a sequence of step functions, hence the development of Bourbaki integration in this problem set serves both as a solid practice ground for convergence of sequences of functions as well as a transition to calculus proper.

¹The redeeming quality of Bourbaki integration theory is that it generalizes fairly straightforwardly to functions taking values in *infinite dimensional spaces*, as opposed to $\mathbb{R}$, which later on proves useful for establishing *remainder formulas for Taylor approximation* in infinite dimensions, among other things. In 3210, we'll not really develop the theory in these directions.

Bourbaki Integration Theory Here are some definitions and notation you'll need for this problem set. Let I be a compact interval, $\sigma \in F(I; \mathbb{R})$. σ is called a **step function** if

- σ attains finitely many values (that is, $\text{im}(\sigma)$ is a finite set), and
- For any $y \in \text{im}(\sigma)$, there are finitely many pairwise disjoint intervals whose union is the level set $\sigma^{-1}(y)$.

A function $f \in F(I; \mathbb{R})$ is called a **Bourbaki integrable function** (or **regulated function**) if it's the limit of a sequence of step functions on I in with respect to the C^0 metric. We denote by $B(I; \mathbb{R})$ the collection of all real valued Bourbaki integrable functions on I .

For any $f \in B(I; \mathbb{R})$, a **(unsigned) Bourbaki integral** (or **regulated integral**) of f is a real number

$$\mathfrak{I}_B(f; \sigma_\bullet) = \lim_{n \rightarrow \infty} \sum_{y \in \text{im}(\sigma_n)} y \left(\sum_{i=1}^m \ell(J_i) \right),$$

provided the limit exists, where

- $\sigma_\bullet$ is a sequence of step functions that converges uniformly to f ,
- for each n , and for each $y \in \text{im}(\sigma_n)$, $J_1, J_2, \dots, J_m \subseteq I$ are disjoint intervals such that $\bigcup_{i=1}^m J_i = \sigma_n^{-1}(y)$, and
- for each i , $\ell(J_i) = \sup(J_i) - \inf(J_i)$ is the length of the interval J_i .

If, *in addition*, the limit defining the number $\mathfrak{I}_B(f; \sigma_\bullet)$ is independent of the approximating sequence $\sigma_\bullet$, then we call this limit **the (unsigned) Bourbaki integral** of f and denote it by

$$\mathfrak{I}_B(f) \quad \text{or} \quad \int_B f(x) dx \quad \text{or} \quad (B) \int_I f(x) dx$$

in case multiple senses of integrals are involved, and by simply

$$\int_I f(x)dx$$

in case it is clear from context that we are only considering Bourbaki integrals.

2 Generative AI and Computer Algebra Systems Regulations

This section applies only if you choose to use either a **generative AI tool** (e.g. chatbots) or a **computer algebra system (CAS)** while working on this problem set. If you do not use such tools, you may skip this section.

The use of these tools while completing this problem set is permitted, provided it is done responsibly and in a manner that supports your learning. Note however that in exams you will not be allowed to use any AI or CAS.

2.1 Disclosure

If you use such tools, you must disclose this fact in the designated section of the form you will complete as part of your submission. You must also share your reflections on which parts of the problem required your own mathematical judgment, insight, or decision-making, and which parts could reasonably be delegated to a computational or generative tool.

2.2 Guidelines for Responsible Use

If you use a generative AI tool or chatbot, you must adhere to the following guidelines:

- During the chat you may share with the chatbot parts or all of this specification document, as well as parts of the textbook or other sources.
- Directly asking the tool for complete problem solutions is prohibited.
- Appropriate uses include asking for conceptual explanations, checking intermediate steps, clarifying definitions or theorems, or exploring alternative solution approaches, provided that the final work submitted is your own.
- Regardless of the tools used, you are expected to understand, justify, and be able to reproduce all submitted work independently. The use of AI or CAS does not reduce or replace this expectation.

3 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details.

When tackling statements in MATH 3210, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.

2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example, you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook. Going forward, and possibly for the rest of this problem set, such correspondences will not be mentioned; it'll be up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. In this problem your job is to investigate **homeomorphisms** of the unit interval $I = [0, 1]$.

- (a) If $f_* : I \rightarrow I$ is a homeomorphism and $\varepsilon \in \mathbb{R}_{>0}$, then there is a piecewise linear homeomorphism $f : I \rightarrow I$ such that $d_{C^0}(f, f_*) < \varepsilon$, where a function $f : I \rightarrow I$ is **piecewise linear** if its graph consists of finitely many line segments².
- (b) If $f_\bullet$ is a sequence of homeomorphisms of I and it converges uniformly to a function $f_\infty : I \rightarrow I$, then f_∞ is a homeomorphism³.
- (c) If f_* is a homeomorphism of I , then there is an $\varepsilon \in \mathbb{R}_{>0}$ such that for any continuous function $f : I \rightarrow I$, if $d_{C^0}(f, f_*) < \varepsilon$, then f is a homeomorphism⁴.
- (d) If $f, g : I \rightarrow I$ are two strictly increasing homeomorphisms, then there is a homeomorphism $\Phi : I \rightarrow I$ such that $\Phi \circ f = g \circ \Phi$ ⁵.

²More succinctly, any homeomorphism can be C^0 -approximated by a PL homeomorphism.

³More succinctly, being a homeomorphism is a **C^0 -closed** property.

⁴More succinctly, any **C^0 -perturbation** of a homeomorphism is a homeomorphism.

⁵More succinctly, any two strictly increasing homeomorphisms are the same up to a **C^0 change of variables** or **topological conjugacy**.

2. In this problem, your job is to establish relationships between Bourbaki integrability and other properties of functions we have encountered so far. Let I be a compact interval.
- (a) Any Bourbaki integrable function on I is bounded.
 - (b) Any nonincreasing or nondecreasing function on I is Bourbaki integrable.
 - (c) Any piecewise continuous function on I is Bourbaki integrable. Here a function is **piecewise continuous** if it is continuous at all but finitely many points in I .
 - (d) Any uniform limit of any sequence of Bourbaki integrable functions on I is Bourbaki integrable.
 - (e) For any $f \in B(I; \mathbb{R})$ and any $\phi \in C^0(\mathbb{R}; \mathbb{R})$, $\phi \circ f \in B(I; \mathbb{R})$.
3. **This problem is a prequel to the next problem.** Let I be a compact interval.
- (a) Let $f \in B(I; \mathbb{R})$ and $\sigma_\bullet$ be a sequence of step functions that uniformly converges to f . Then the limit defining $\mathfrak{I}_B(f; \sigma_\bullet)$ exists.
 - (b) Let $f \in B(I; \mathbb{R})$ and $\sigma_\bullet, \bar{\sigma}_\bullet$ be two sequences of step functions that uniformly converge to f . Then $\mathfrak{I}_B(f; \sigma_\bullet) = \mathfrak{I}_B(f; \bar{\sigma}_\bullet)$.
4. **This problem is a sequel to the previous problem.** Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow B(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$. Consider the following statements:
- I: $f_\bullet \rightarrow f_\infty$ uniformly.
 - II: $f_\bullet \rightarrow f_\infty$ pointwise.
 - III: f_∞ is Bourbaki integrable.

IV: f_∞ is Bourbaki integrable and

$$\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx.$$

V: For each $n \in \mathbb{Z}_{\geq 0}$, $|f_n - f_\infty|$ is Bourbaki integrable and

$$\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx = 0.$$

Then

- (a) If **I**, then **IV** and **V**.
- (b) If **II**, then **III**.
- (c) If **II** and **III**, then **IV**.
- (d) If **II** and **IV**, then **V**.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/bof9ozbFLdd8NGuc6>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

<https://www.gradescope.com/courses/1212058/assignments/7390770>,

see the Gradescope [documentation](#) for instructions.

5 When to Submit

This problem set is due on February 20, 2026 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.